calm_weighted_matched

equity strategy | family: equity-volatility | status: validated | derived from `calm_weighted`

What this strategy does

Calm Weighted resized so its ups and downs match Balanced Leverage exactly (4.94% a year)

Why it should work. PEAD is a stronger, cleaner signal in low-vol stocks; size positions by inverse risk.

Design notes (from the strategy's source file).

Calm Weighted -- Balanced Leverage, with bigger bets on calmer stocks (sized by inverse idiosyncratic volatility).

Within every (quarter, side, sector, size) cell, each event's weight is scaled by 1/ivol, so calm stocks get more capital than wild ones. Rationale: the earnings drift is worth more per unit of risk in low-vol names (the SUE x vol interaction is t = -2.4 to -2.7 in Fama-MacBeth), and the engine's cost/liquidity model charges the same bps regardless of vol. Raises Sharpe 1.30 -> 1.47 with no fitted parameter, in all three sub-eras.

`CalmWeightedMatched` re-sizes it to Balanced Leverage's exact volatility to answer "at the SAME risk, how much more return": 7.31% vs 6.49%.

parameter	value
base_unit_fraction	0.003468808347716693
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.0

Performance

metric	calm weighted matched	market (CRSP VW)
Annualized return	5.83%	9.94%
Annualized volatility	4.88%	19.35%
Sharpe (NAV-pct)	1.18	0.59
Sortino	1.72	
Max drawdown	-18.7%	-55.5%
Calmar	0.31	
Final NAV ($10M start)	$30,129,519	
Worst day	-3.72%	
% positive days	50.8%	
Sharpe 1996-2001	1.94	
Sharpe 2002-2007	1.47	

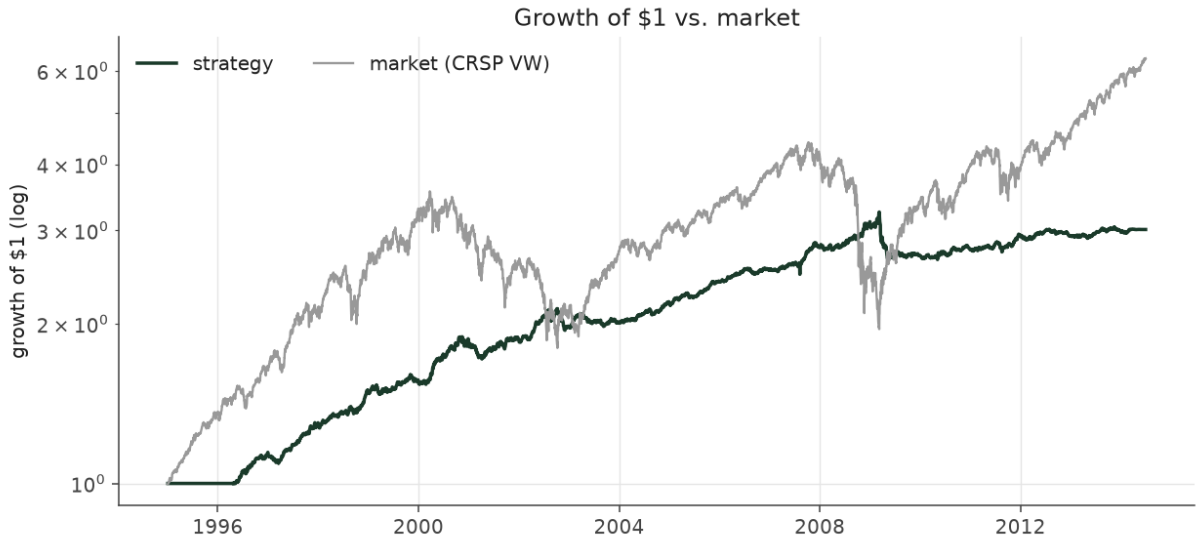
Sharpe 2008-2013	0.25	

Correlation of daily returns with the market: **-0.05**.

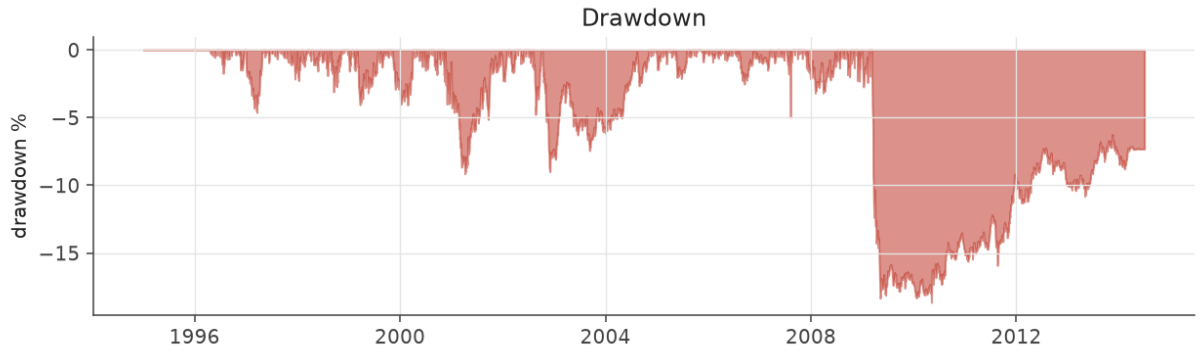
• best Sharpe among equity: `calm_rank_tilt` (1.81); this one: 1.18

• best return among equity: `market_boost_max` (0.15); this one: 0.0583

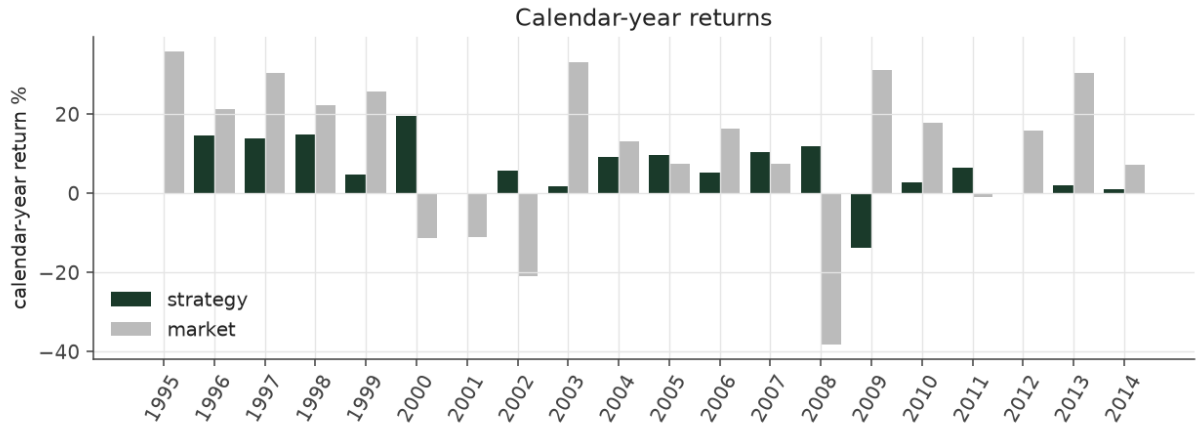
• best Calmar among equity: `calm_rank_tilt` (0.614); this one: 0.312



NAV (log scale)



Drawdown from running peak

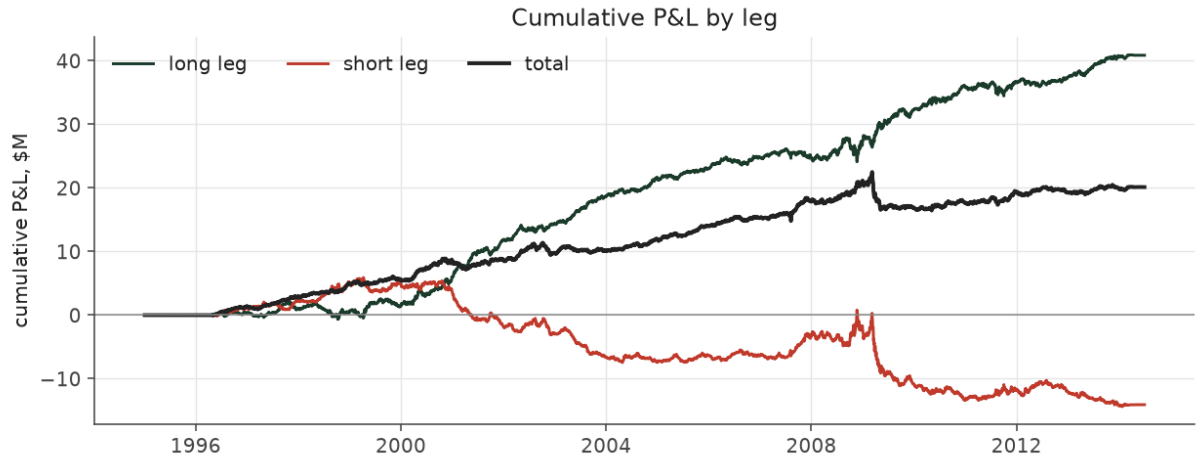


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	40.86	-14.09
P&L after costs ($M)	37.64	-17.51
of which common (average-event) component ($M)	27.18	-28.06

of which excess / PEAD component ($M)	13.68	13.97
excess minus costs ($M)	10.46	10.55
gross return per $ traded (bp)	249.47	-82.88
positions in the money (hit rate)	0.56	0.48
stand-alone Sharpe (leg only)	1.28	-0.34

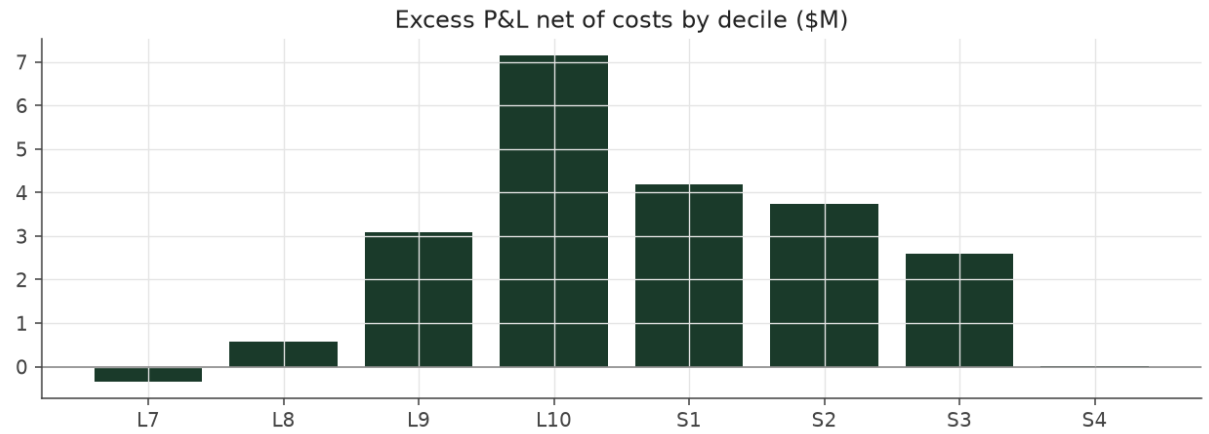
Daily P&L correlation between the two legs: **-0.78**.

• **The short leg loses money in absolute terms** (-14.1M before costs) while the long leg earns +40.9M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 27.2M / 28.1M. Stripping that common component, the surprise signal itself earned +13.7M on the long side and +14.0M on the short side.

• The two legs' daily P&L are -0.78 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.28 long / -0.34 short vs 1.18 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	4,382	99.34	1.37	-0.15	0.20	-0.35	-1.7%	-35.33
long	8	12,698	351.01	7.15	1.27	0.69	0.58	2.8%	16.60
long	9	15,395	504.72	10.39	4.06	0.98	3.08	14.7%	61.02
long	10	17,762	682.83	18.73	8.49	1.34	7.15	34.0%	104.67
short	1	16,242	681.39	-7.48	5.62	1.42	4.20	20.0%	61.57
short	2	16,137	587.36	-6.10	4.91	1.17	3.75	17.8%	63.81
short	3	11,827	385.71	-3.31	3.35	0.75	2.59	12.3%	67.25
short	4	1,783	45.18	-0.62	0.09	0.08	0.01	0.0%	1.82



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	10,937	270.71	0.21	1.82	1.02	0.80	3.8%	29.55
2	18,554	610.52	6.24	8.41	1.72	6.70	31.9%	109.74

3	20,808	768.40	6.05	7.78	1.63	6.14	29.2%	79.95
4	22,405	836.51	4.76	5.95	1.33	4.61	22.0%	55.13
5	23,522	851.40	2.86	3.69	0.94	2.75	13.1%	32.29

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	10,124	191.34	2.01	3.32	0.96	2.37	11.3%	123.74
2	19,346	681.98	4.22	7.24	2.05	5.19	24.7%	76.15
3	21,276	847.91	6.73	8.81	1.70	7.11	33.9%	83.87
4	22,262	821.24	4.50	5.17	1.15	4.02	19.1%	48.97
5	23,214	794.76	2.60	3.03	0.80	2.23	10.6%	28.12

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	33,598	1,151.13	0.93	10.23	2.16	8.07	38.4%	70.11
41-60d	23,226	843.91	3.21	6.82	1.83	4.99	23.8%	59.15
61-90d	31,609	1,092.13	10.22	5.71	1.98	3.73	17.7%	34.12
>90d	7,793	250.37	5.78	4.89	0.68	4.21	20.1%	168.37

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	30,165	719.54	8.87	10.63	1.41	9.22	43.9%	128.15
2002-07	31,557	1,151.13	8.81	10.98	2.25	8.73	41.6%	75.85
2008-13	34,504	1,466.88	2.44	6.04	2.99	3.05	14.5%	20.81

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	16,166	297.04	0.16	1.09	0.62	0.47	2.2%	15.81
Chems	2,514	236.77	1.06	1.46	0.41	1.05	5.0%	44.24
Durbl	2,755	255.23	1.66	2.27	0.46	1.81	8.6%	70.87
Enrgy	5,919	284.83	3.21	3.80	0.53	3.27	15.6%	114.73
Hlth	9,694	321.37	1.47	2.12	0.66	1.46	7.0%	45.47
Manuf	10,313	299.95	1.61	2.35	0.63	1.72	8.2%	57.39
Money	11,832	245.92	2.06	2.60	0.57	2.02	9.6%	82.28
NoDur	4,763	287.47	1.43	1.92	0.54	1.38	6.6%	48.04
Other	13,276	281.24	2.20	2.90	0.62	2.27	10.8%	80.80
Shops	10,682	321.89	3.01	3.65	0.66	2.98	14.2%	92.71

Telcm	3,863	283.65	2.06	2.62	0.52	2.10	10.0%	74.22
Utils	4,449	222.20	0.22	0.87	0.41	0.46	2.2%	20.79

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	29,615	783.87	1.08	3.52	1.42	2.11	10.0%	26.88
low vol	34,228	1,417.19	10.67	14.12	3.01	11.11	52.9%	78.40
mid vol	30,825	1,084.48	7.52	9.04	2.10	6.94	33.0%	63.97

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	5.83%	0.00%
0.5%	1.7	5.51%	0.31%
1.0%	3.4	5.18%	0.65%
2.0%	6.8	4.45%	1.38%
5.0%	17.0	1.41%	4.41%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.